

ILE DE GROIX, France

WMO: 072030

Lat: 47.652N

Lon: 3.502W

Elev: 54

StdP: 100.68

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.7	1.9	-4.0	2.7	2.5	-2.3	3.1	3.1	28.1	8.5	25.4	8.4	7.7	70	0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	25.7	19.1	23.8	18.3	22.3	17.8	19.9	24.0	19.1	22.3	18.5	21.2	5.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	21.0	18.0	13.0	20.4	17.4	12.5	19.8	57.1	24.3	54.5	22.4	52.6	21.2	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	18.7	15.7	DB	-0.9	29.7	2.0	2.6	-2.4	31.6	-3.5	33.2	-4.7	34.7	-6.1	36.6	(4)
(5)			WB	-2.5	21.6	1.9	1.5	-3.8	22.7	-4.9	23.6	-5.9	24.4	-7.3	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	8.0	8.0	9.4	11.4	14.1	16.7	18.2	18.1	17.2	14.5	11.4	9.0	(6)	
	DBStd	4.44	2.75	2.47	2.33	2.22	2.25	2.40	2.31	1.74	1.97	2.10	2.38	2.94	(7)	
	HDD10.0	247	71	62	37	9	1	0	0	0	0	1	13	54	(8)	
	HDD18.3	2013	321	290	277	208	134	62	31	25	47	119	209	291	(9)	
	CDD10.0	1344	8	5	18	51	126	200	254	250	214	141	54	22	(10)	
	CDD18.3	68	0	0	0	0	2	12	26	17	11	1	0	0	(11)	
	CDH23.3	224	0	0	0	0	8	53	89	48	26	1	0	0	(12)	
(13)	CDH26.7	41	0	0	0	0	1	13	17	8	2	0	0	0	(13)	
(14) Wind	WSAvg	7.6	8.6	8.6	7.9	7.1	6.9	6.8	6.9	6.6	6.6	7.3	8.6	8.8	(14)	
(15) Precipitation	PrecAvg	842	103	77	58	64	56	45	48	47	57	93	98	97	(15)	
	PrecMax	1062	208	176	189	162	132	97	107	107	146	174	202	194	(16)	
	PrecMin	604	17	15	6	15	12	6	6	8	5	18	25	20	(17)	
	PrecStd	111	54	43	39	39	27	25	23	27	39	37	46	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	13.6	18.6	21.5	24.0	28.6	28.7	27.7	26.1	20.7	16.9	14.4	(19)	
		MCWB	12.2	11.2	12.0	15.1	17.4	20.2	20.0	20.2	18.7	17.5	14.8	12.8	(20)	
	2%	DB	12.6	12.1	15.1	18.5	21.2	24.4	25.6	24.3	23.5	19.1	15.9	13.7	(21)	
		MCWB	11.6	10.6	11.2	13.5	16.2	18.4	19.2	18.7	17.9	16.4	14.6	12.4	(22)	
	5%	DB	12.0	11.6	13.2	16.2	19.2	22.4	23.7	22.1	21.7	18.0	15.2	13.2	(23)	
		MCWB	11.1	10.3	10.5	12.5	15.4	17.7	18.7	18.0	17.3	15.8	13.8	12.0	(24)	
	10%	DB	11.5	11.0	12.1	14.6	17.5	20.5	21.9	20.8	20.3	17.2	14.5	12.7	(25)	
		MCWB	10.5	10.0	10.2	11.6	14.5	16.9	18.1	17.4	16.7	15.4	13.0	11.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	11.8	12.9	15.8	18.2	21.3	21.1	20.7	19.9	18.1	15.9	13.5	(27)	
		MCDB	12.8	12.7	16.9	19.9	22.4	26.3	26.8	25.7	24.1	20.0	16.3	13.9	(28)	
	2%	WB	12.0	11.2	11.9	14.0	16.8	19.3	19.9	19.5	18.9	17.2	14.8	13.0	(29)	
		MCDB	12.4	11.7	13.7	17.1	20.0	23.3	24.0	22.7	21.6	18.4	15.5	13.5	(30)	
	5%	WB	11.4	10.7	11.3	13.0	15.8	18.2	19.1	18.7	18.3	16.5	14.2	12.4	(31)	
		MCDB	11.8	11.3	12.5	15.4	18.5	21.5	22.4	21.1	20.5	17.5	15.0	13.0	(32)	
	10%	WB	10.8	10.3	10.7	12.2	14.9	17.2	18.5	18.1	17.6	15.8	13.3	11.8	(33)	
		MCDB	11.3	10.9	11.7	14.1	17.1	19.9	21.3	20.0	19.5	16.8	14.3	12.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.4	4.0	4.9	5.5	5.5	5.8	5.6	5.6	6.0	4.5	3.5	3.3	(35)	
		MCDBR	2.9	4.2	6.8	8.3	8.4	9.4	9.4	8.0	8.2	5.4	3.6	2.8	(36)	
	5% WB	MCWBR	3.2	3.6	4.2	4.8	4.9	5.1	4.8	4.3	4.2	3.6	3.5	3.1	(37)	
		MCWBR	2.8	3.4	5.3	6.8	7.3	8.3	7.5	6.5	6.5	4.5	3.4	2.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.334	0.347	0.372	0.384	0.389	0.386	0.383	0.373	0.361	0.360	0.339	0.327	(40)	
		taud	2.425	2.407	2.352	2.336	2.368	2.402	2.411	2.458	2.473	2.465	2.445	2.402	(41)	
	Ebn at Noon	Ebn at Noon	737	810	843	868	876	877	875	870	849	786	725	692	(42)	
		Edh at Noon	69	85	105	117	117	114	112	102	92	80	67	63	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.05	1.97	3.16	4.75	5.62	6.17	5.90	5.13	4.08	2.33	1.33	0.92	(44)	
	RadStd	RadStd	0.10	0.19	0.32	0.46	0.38	0.56	0.27	0.34	0.28	0.19	0.11	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page